

Assessing 5-FU + levamisole treatment efficacy in colorectal cancer - presented Wednesday at 11:15 am

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Abstract

Purpose We aim to assess prognostic factors for CRC and efficacy of three treatments: surgery alone, levamisole alone, and levamisole + 5-FU.

Methods Three hundred and sixty three patients participated in the randomized controlled trial. There were 125 patients in the control group, 120 in levamisole group, and 118 in the levamisole + 5-FU group. A Cox model was fit to examine prognostic factors. Progression-free survival (PFS) and overall survival (OS) times were calculated with Kaplan-Meier methods.

Results Levamisole + 5-FU had higher OS and PFS rates as compared to levamisole alone.

Conclusion Levamisole + 5-FU is an effective treatment for CRC (HR = 0.68 (0.45,1.02), $p = 0.059$).

Introduction

Worldwide, colorectal cancer (CRC) is the second most common cause of cancer-related deaths. In 2020, approximately 1.9 million people were diagnosed with CRC and 915,000 patients died from CRC (Yu et al., 2023). Treatment outcomes after resection are excellent. Adjuvant chemotherapy, such as 5-FU can improve both disease-free and overall survival for patients with stage III CRC (Yu et al., 2023). Postoperative adjuvant chemotherapy is a controversial issue for stage II CRC. Perforation, obstruction, and poor differentiation are all high-risk factors for stage II CRC. As many as 40-50% of patients with locoregional lymph node metastasis at diagnosis relapse when treated with surgery alone (Akdeniz et al., 2021). The objective of this project is to assess prognostic factors for CRC and efficacy of surgery alone, levamisole, and 5-FU + levamisole.

Methods

Study population

Three hundred and sixty-three patients participated in the study and were randomized to surgery alone, levamisole + 5-FU, or levamisole only.

Statistical analysis

Statistical analyses were performed using R (version 4.5.2, R Core Team, 2025). Frequencies and percentages were reported for categorical variables. Median, min, and max were reported for age. An α level of < 0.05 was considered statistically significant. Kaplan-Meier methods were used to assess overall survival (OS) and progression-free survival (PFS). OS was defined as time in years since registration until death or last known contact date. PFS was defined time from registration until progression/recurrence. Log-rank test was used to assess differences between treatment groups. A full Cox model was fit

with all available predictors and a reduced model was fit using predictors from the full model with $p < 0.05$. A Likelihood Ratio Test was used to compare full and reduced models. Hazard ratios (HR) and 95% confidence intervals (CI) from the final model were calculated.

Results

Three hundred sixty-three patients enrolled in the study. Among them, 125 patients were in the control group, 120 patients were in levamisole only group, and 118 patients were in the levamisole+5-FU group. Median age at diagnosis was 61 years for the entire population. Median age for control, levamisole, and levamisole+5-FU groups were 62,61,and 61 years, respectively. Table 1 contains a summary of patient characteristics.

Survival

Sixty (48.0%) patients in the control group, 49 (40.8%) patients in levamisole only group, and 43 (36.4%) patients in levamisole+5-FU experienced relapse. Median OS was 3.5 years in control group, not estimable in levamisole group, and not estimable in 5-FU + levamisole group. Median PFS was 2.9 years in control group, and not estimable in levamisole and levamisole + 5-FU groups. The 2-year and 4-year PFS rates were 68% and 59%, respectively, for 5-FU + levamisole (Table 4). The 2-year and 4-year OS rates were 82% and 72%, respectively, for 5-FU + levamisole. Both PFS and OS 2-year and 4-year survival rates were higher for 5-FU + levamisole group than the other two groups. A marginally significant difference ($p = 0.052$) was observed between the three groups for PFS. Similarly, a statistically significant difference was found between the three groups for OS ($p = 0.043$).

A full Cox proportional hazards model was fit with all predictors for PFS. Predictors were kept in the model if the p-value was < 0.05 . A reduced model was then fit. The full and reduced models were compared with LRT. The LRT statistic was 31.996 ($p = 0.059$) so the reduced model was used. The reduced model included treatment, extent of spread, primary site, histology type, nodes, and differentiation. The HR for 5-FU + levamisole vs. placebo was 0.68, indicating that the hazard of relapse is 32% lower in patients taking 5-FU + levamisole vs placebo when holding all other predictors constant. The HR for levamisole only is 0.81, indicating that the hazard of relapse is 19% lower in patients taking levamisole only vs placebo when holding all other predictors constant. Nodal involvement was a highly significant prognostic factor in the model ($p < .001$). Results from the reduced Cox model are shown in Table 4. PH assumption was checked by plotting Schoenfeld residuals.

Discussion

We compared three different treatment types for CRC. Although the p-value was only marginally significant in the Cox model, the HR suggests that 5-FU + levamisole was the best treatment regimen. Additionally, the 5-FU + levamisole group had higher rates of survival for both OS and PFS. However, this data was quite old. Future research should focus on later treatments that have been developed in the time since this data was collected.

Table 1: Patient Characteristics

Characteristic	Control (N = 125)	Levamisole only (N = 120)	5-FU + Levamisole (N = 118)
Age (Years)			
Median	62	61	61
Min, Max	24, 84	28, 83	27, 81
Sex			
Male	64 (51%)	66 (55%)	57 (48%)
Female	61 (49%)	54 (45%)	61 (52%)
Extent of local spread			
Submucosa/not muscle	2 (1.6%)	1 (0.8%)	6 (5.1%)
Muscular/not serosa	16 (13%)	15 (13%)	8 (6.8%)
Serosa/not contiguous	99 (79%)	103 (86%)	98 (83%)
Contiguous structures	8 (6.4%)	1 (0.8%)	6 (5.1%)
Nodal involvement			
1-4	87 (70%)	84 (70%)	84 (71%)
>4	38 (30%)	36 (30%)	34 (29%)
Location of primary neoplasm			
Cecum neoplasm	8 (6.4%)	9 (7.5%)	14 (12%)
Right colon	26 (21%)	24 (20%)	33 (28%)
Hepatic flexure	17 (14%)	18 (15%)	11 (9.3%)
Transverse colon	7 (5.6%)	9 (7.5%)	4 (3.4%)
Splenic flexure	8 (6.4%)	8 (6.7%)	9 (7.6%)
Left colon	5 (4.0%)	6 (5.0%)	5 (4.2%)
Sigmoid colon	7 (5.6%)	9 (7.5%)	6 (5.1%)
Rectosigmoid	41 (33%)	35 (29%)	33 (28%)
Rectum	0 (0%)	0 (0%)	0 (0%)
Multiple sites	6 (4.8%)	2 (1.7%)	3 (2.5%)
Histologic type			
Adenocarcinoma	115 (92%)	110 (92%)	101 (86%)
Colloid(mucinous)	7 (5.6%)	8 (6.7%)	15 (13%)
Signet ring type	2 (1.6%)	1 (0.8%)	2 (1.7%)
Other	1 (0.8%)	1 (0.8%)	0 (0%)
Differentiation			
Well	12 (9.9%)	17 (15%)	17 (15%)
Moderate well (gr 2-3)	85 (70%)	79 (69%)	79 (68%)
Poor (grade 4)	24 (20%)	19 (17%)	20 (17%)
Unknown	4	5	2
Initiation of therapy			
7-20 days after surgery	40 (32%)	35 (29%)	32 (27%)

21-30 days after surgery	85 (68%)	85 (71%)	86 (73%)
Adherence			
Yes	14 (11%)	14 (12%)	10 (8.5%)
No	111 (89%)	106 (88%)	108 (92%)
Regional implants			
Yes	7 (5.6%)	6 (5.0%)	12 (10%)
No	118 (94%)	114 (95%)	106 (90%)
Organ involvement			
Yes	16 (13%)	13 (11%)	7 (5.9%)
No	109 (87%)	107 (89%)	111 (94%)
Serosal involvement			
Yes	102 (82%)	103 (86%)	100 (85%)
No	23 (18%)	17 (14%)	18 (15%)
Obstruction			
Yes	28 (22%)	20 (17%)	18 (15%)
No	97 (78%)	100 (83%)	100 (85%)
Perforation			
Yes	4 (3.2%)	5 (4.2%)	4 (3.4%)
No	121 (97%)	115 (96%)	114 (97%)

¹ 5-FU=fluorouracil

Fig. 1 K-M curve of PFS

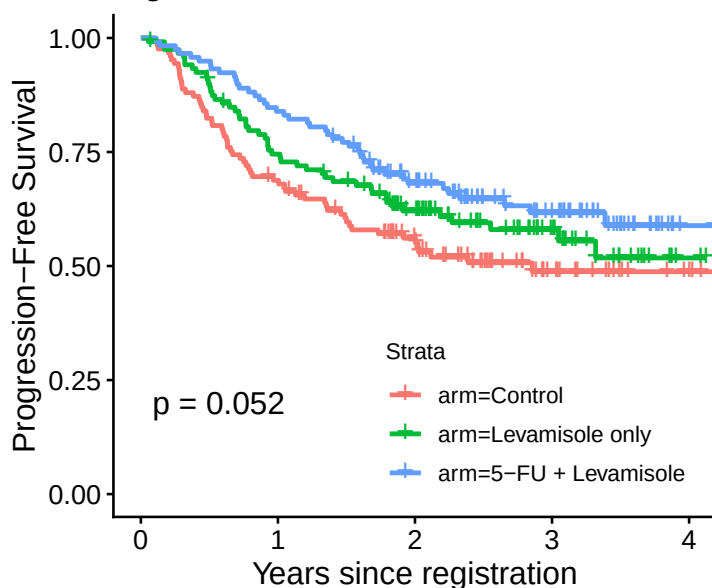


Fig. 2 K-M curve of overall survival

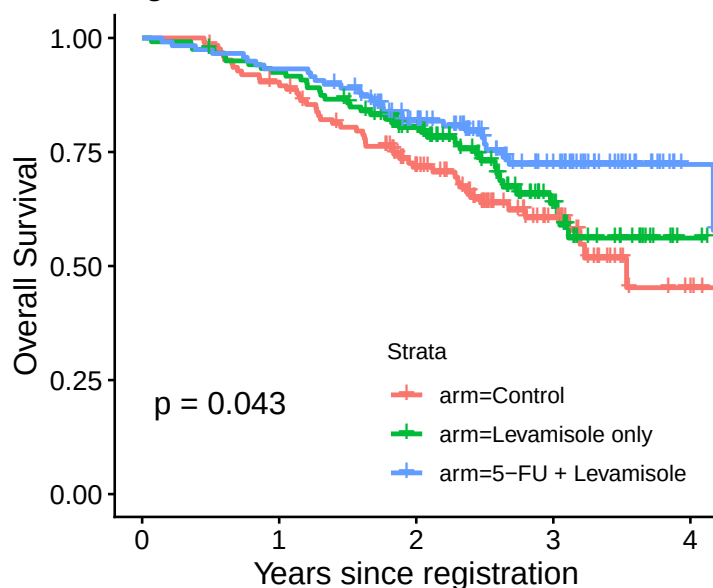


Table 2: 2-year and 4-year survival for OS and RFS

Factor	RFS			OS		
	Event N	2 Year RFS (%)	4 Year RFS (%)	Event N	2 Year OS (%)	4 Year OS (%)
Treatment Regimen	152			114		
Control		56% (48%, 66%)	49% (40%, 59%)		72% (64%, 80%)	45% (32%, 65%)
Levamisole only		62% (54%, 72%)	52% (41%, 65%)		80% (73%, 88%)	56% (45%, 70%)
5-FU + Levamisole		68% (60%, 77%)	59% (49%, 70%)		82% (75%, 89%)	72% (64%, 82%)

Table 3: LRT Results Comparing Cox Full vs. Cox Reduced

loglik	Chisq	Df	Pr(> Chi)
-783.817	NA	NA	NA
-799.815	31.996	21	0.059

Table 4: Reduced Cox Model Regression Results

Factor	HR	95% CI	p-value
Treatment group			
Control	—	—	
Levamisole only	0.81	0.55, 1.19	0.3
Levamisole + 5-FU	0.68	0.45, 1.02	0.059
Histologic type			
Adenocarcinoma	—	—	
Colloid(mucinous)	0.84	0.42, 1.65	0.6
Signet ring type	0.55	0.08, 4.00	0.6
Other	7.96	1.92, 33.0	0.004
Nodal involvement			
1-4	—	—	
>4	2.04	1.47, 2.84	<0.001

Abbreviations: CI = Confidence Interval, HR = Hazard Ratio

References

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- Yu, Y.-L., Tseng, W.-K., Liao, C.-K., Yeh, C.-Y., Chen, H.-H., Liu, Y.-H., Liaw, Y.-W., & Fan, C.-W. (2023). Using oral tegafur/uracil (Uft) plus leucovorin as adjuvant chemotherapy in stage II colorectal cancer: A propensity score matching study from Taiwan. *BMC Cancer*, 23(1), 900. <https://doi.org/10.1186/s12885-023-11310-6>